

Question description

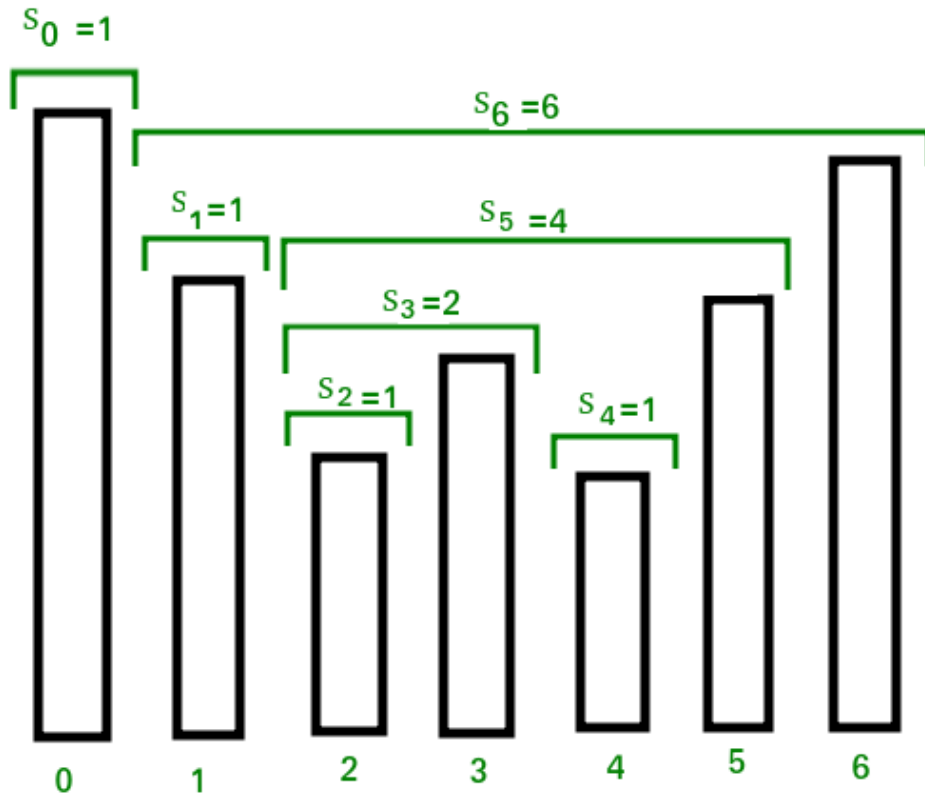
The **stock span problem** is a financial problem where we have a series of n daily price quotes for a stock and we need to calculate span of stock's price for all n days.

The span S_i of the stock's price on a given day i is defined as the maximum number of consecutive days just before the given day, for which the price of the stock on the current day is less than or equal to its price on the given day.

For example, if an array of 7 days prices is given as $\{100, 80, 60, 70, 60, 75, 85\}$,

then the span values for corresponding 7 days are $\{1, 1, 1, 2, 1, 4, 6\}$

Question No:6



Constraints

$0 < n < 1000$

$0 < \text{price}[i] < 1000000$

Input Format:

First line indicates the number of days

second line indicates the price quoted for above mentioned days

Output Format:

Single line represents the span values for corresponding days

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

6
9 8 7 6 5 4

EXPECTED OUTPUT

1 1 1 1 1 1

Test Case 2

INPUT (STDIN)

6
10 41 15 90 10 70

EXPECTED OUTPUT

1 2 1 4 1 2

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void printArray(int
arr[],int n)

Test Case 2

KEYWORD

void calculateSpan(int
price[],int n,int S[])

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

220

Test Case 3

NLOC

35

Code:

```
1  #include <stdio.h>
2
3  #define MAX 100000
4
5  int stack[MAX];
6  int top = -1;
7
8  int main()
9  {
10     int n;
11     int price[MAX];
12     int span[MAX];
13
14     scanf("%d", &n);
15
16     for (int i = 0; i < n; i++)
17     {
18         scanf("%d", &price[i]);
19     }
20
21     for (int i = 0; i < n; i++)
22     {
23         /* Remove all previous prices <= current price */
24         while (top != -1 && price[stack[top]] <= price[i])
25         {
26             top--;
27         }
28
29         /* Calculate span */
30         if (top == -1)
31         {
32             span[i] = i + 1;
33         }
34         else
35         {
36             span[i] = i - stack[top];
37         }
38     }
39 }
```

```
38
39      /* Push current day's index */
40      stack[++top] = i;
41  }
42
43      /* Print span values */
44      for (int i = 0; i < n; i++)
45      {
46          printf("%d", span[i]);
47
48          if (i < n - 1)
49          {
50              printf(" ");
51          }
52      }
53
54      return 0;
55 }
```

Output:

Test Case-1:

```
6
9 8 7 6 5 4
1 1 1 1 1 1
-----
Process exited after 62.3 seconds with return value 0
Press any key to continue . . . |
```

Test Case-2

```
6
10 41 15 90 10 70
1 2 1 4 1 2
-----
Process exited after 28.83 seconds with return value 0
Press any key to continue . . . |
```

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